

Response & Bode Diagram: Exercise

The solution of the exercises proposed in this document was given during class.

Exercise 1)

Consider the following systems:

$$\begin{aligned} G_1(s) &= \frac{10}{s+1} & G_3(s) &= \frac{10}{s^2 + 11s + 10} \\ G_2(s) &= \frac{s-10}{s+1} & G_4(s) &= \frac{10}{s^2 + 15s + 100} \end{aligned}$$

and recalling the following:

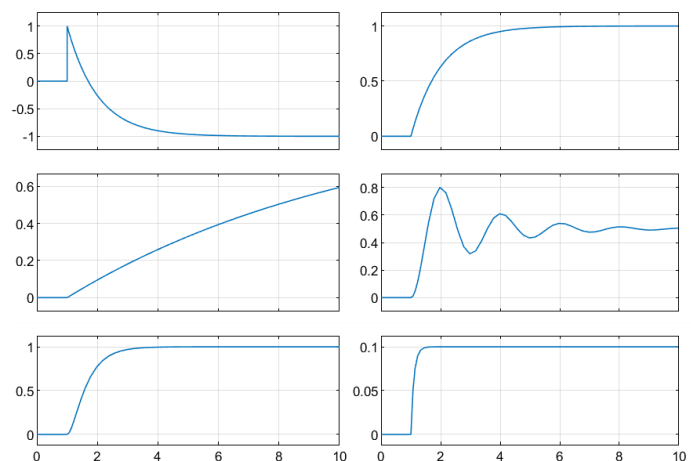
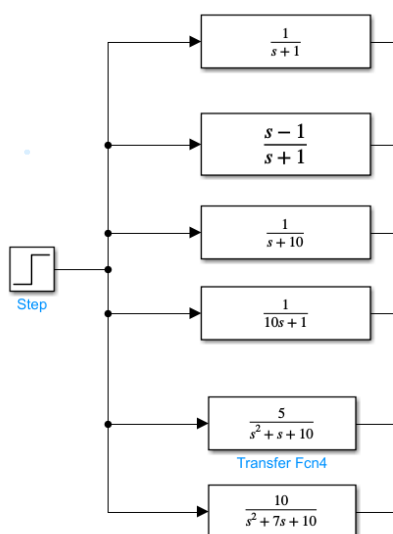
<i>1st Order Strictly Proper</i>				<i>1st Order Proper</i>	<i>2nd Order: complex pole</i>	
T_s	T_r	T_{a5}	T_{a1}	$T_{a\varepsilon}$	$S\%$	$T_{a\varepsilon}$
$\approx 2.2T$	$\approx 0.7T$	$\approx 3T$	$\approx 4.7T$	$T \ln \frac{ 1-\alpha }{0.01\varepsilon}$	$100e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}}$	$-\frac{\ln 0.01\varepsilon}{\xi\omega_n}$

determine the following indexes from a unitary step response

- Equilibrium value
- Settling time at 5%
- Overshooting percentage

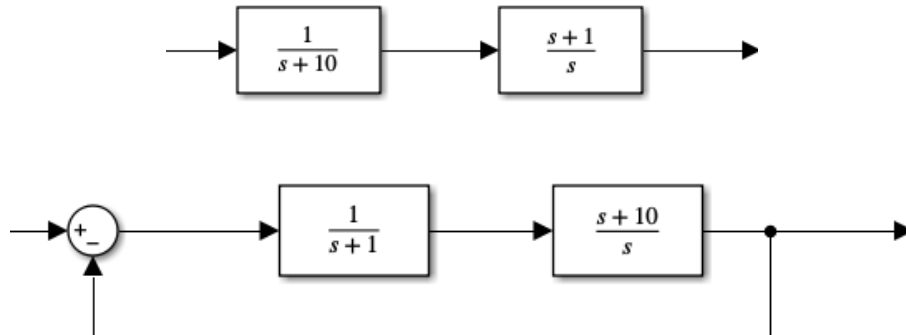
Exercise 2)

Connect the systems with the correct step response



Exercise 3)

Given the systems



1. evaluate the response to the following signals:
 - a. $10\sin(10t)$
 - b. $\sin(100t)+3\cos(t)$
2. Define the Bode Diagram

Exercise 1)

Consider the following systems:

$$G_1(s) = \frac{10}{s+1} \quad G_3(s) = \frac{10}{s^2 + 11s + 10}$$

$$G_2(s) = \frac{s-10}{s+1} \quad G_4(s) = \frac{10}{s^2 + 15s + 100}$$

and recalling the following:

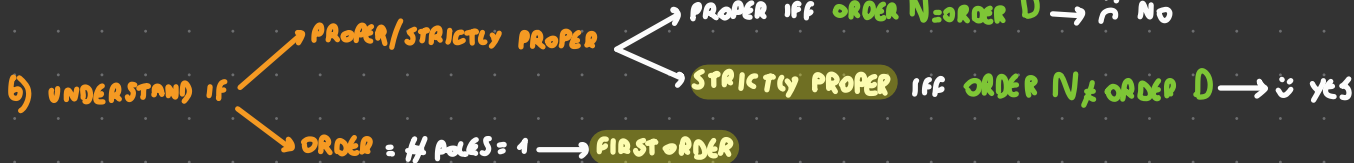
1 st Order Strictly Proper				1 st Order Proper	2 nd Order: complex pole	
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determine the following indexes from a unitary step response

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a) CHECK STABILITY

$$\hookrightarrow \text{POLE}(P): (s+1)=0 \rightarrow s=-1 \rightarrow \text{STABLE } (P: \{-1\} < 0); \quad Z = \{\emptyset\}$$



FIND EQ. VALUES

SINCE STABLE I CAN USE FINAL VALUE THEOREM (FVT)

$$U = \lim_{s \rightarrow 0} s \cdot G(s) \cdot \frac{1}{s} = \lim_{s \rightarrow 0} G(s) = G(s) = 10$$

SINCE UNITARY
SINCE STEP

SETTLING TIME AT 5%

I KNOW THAT IS 1st ORDER AND STRICTLY PROPER, SO:

$$T_{a5} \approx 3T \approx 3 \left| \frac{1}{p} \right| \approx 3 \left| \frac{1}{-1} \right| \approx 3 \text{ SECONDS}$$

POLES

! TUA BISOGNA USARE SI E NON =

OVERSHOOTING %



$$G_1(s) = \frac{10}{s+1}$$

$$G_2(s) = \frac{s-10}{s+1}$$

$$G_3(s) = \frac{10}{s^2 + 11s + 10}$$

$$G_4(s) = \frac{10}{s^2 + 15s + 100}$$

$$G_2(s) = \frac{s-10}{s+1}$$

• CHECK STABILITY

$P \Rightarrow s+1=0 \rightarrow s=-1$ $P = \{-1\}$ THERE IS ONLY ONE POLE $\rightarrow$ SO STABLE ; $Z = \{10\}$

• PROPER/STRICTLY PROPER?

- PROPER: IFF ORDER N = ORDER D $\rightarrow 1=1 \checkmark$ PROPER
- STRICTLY PROPER: IFF ORDER N $\neq$ ORDER D $\rightarrow \checkmark$

• ORDER

- ORDER = # POLES = 1 $\rightarrow$ FIRST ORDER

STABLE, PROPER, 1st ORDER

FIND EG VALUES

- SINCE STABLE I USE FINAL VALUE THEOREM

$$N = \lim_{s \rightarrow 0} s \cdot G(s) \cdot \frac{1}{s} = \lim_{s \rightarrow 0} G(s) = G(0) = -10$$

$\uparrow$ SINCE UNITARY
 $\downarrow$ SINCE STEP

SETTLING TIME AT 5%.

$$T_{d\epsilon} \approx T_{LN} \frac{|1-\alpha|}{0.01\epsilon} \approx 1 \cdot LN \frac{1-1/10}{0.01 \cdot 5} = 2.39 \text{ s}$$

SECONDS

$$T = \left| \frac{1}{p} \right| = \left| \frac{1}{-1} \right| = 1$$

$$\gamma = \frac{1}{2} = \frac{1}{10}$$

$$\alpha = \frac{\gamma}{T} = \frac{1/10}{1} = \frac{1}{10}$$

$$\epsilon = 5$$

$\uparrow$
 PERCENTAGE RANGE (GIVEN FROM EXERCISE)

OVERSHOOTING %.

IFF $\alpha > 1 \rightarrow$ YES 0.1 $\rightarrow$ NO OVERSHOOT BUT RISE

$$G(s) = \frac{10}{s^2 + 11s + 10}$$

CHECK STABILITY

$$P = \text{POLES} \rightarrow s^2 + 11s + 10 = 0 \quad s_1 = -10, s_2 = -1 \quad \text{STABLE}$$

$$P = \{-10, -1\}$$

$$Z = \{\emptyset\}$$

SINCE ORDER $N = 0 \neq \text{ORDER } D = 2 \rightarrow \text{STRICTLY PROPER}$

SYSTEM IS 2nd ORDER

$\Rightarrow$ 2nd ORDER, STRICTLY PROPER, STABLE

EQUILIBRIUM VALUE

SINCE STABLE USE FINAL VALUE THEOREM

$$N = \lim_{s \rightarrow 0} s G(s) \frac{1}{s} = \lim_{s \rightarrow 0} G(s) = 1$$

SETTLING TIME AT 5%

SINCE NO SETTLING TIME FOR THIS SYSTEM

WE USE DOMINANT POLE APPROXIMATION

DOMINANT POLE IS THE ONE CLOSEST TO IMAGINARY AXES $\rightarrow$ OR THE ONE FOR WHICH

$$T = \left| \frac{1}{p} \right| \text{ IS BIGGEST}$$

$$T_1 = \left| \frac{1}{p_1} \right| = \left| \frac{1}{-1} \right| = 1 \quad \leftarrow T_1 < T_2, \text{ SO I USE } p_1 = -1$$

$$T_2 = \left| \frac{1}{p_2} \right| = \left| \frac{1}{-10} \right| = 0.1$$

$$G(s) = \frac{10}{(s+1)(s+10)} \approx \frac{1}{s+1}$$

$$T_{95} \approx 3T \approx 3 \left| \frac{1}{p} \right| = 3 \text{ SECONDS}$$

③ OVERSHOOTING PERCENTAGE

SINCE APPROXIMATED TO 1ST ORDER PROPER $\rightarrow$ NO OVERSHOOT

$$G(s) = \frac{10}{s^2 + 15s + 100}$$

STABILITY

$$P = \text{POLES} \rightarrow s^2 + 15s + 100 = 0$$

$$s_{1,2} = \frac{-15 \pm \sqrt{225 - 4(100)}}{2} = \frac{-15 \pm 5\sqrt{7}}{2}$$

$$PO \rightarrow \text{STABLE since } \operatorname{Re}(s_1) < 0 \text{ and } \operatorname{Re}(s_2) < 0$$

2nd ORDER {2 POLES}

SINCE ORDER $N \neq$ ORDER $D \rightarrow$ STRICTLY PROPER

EQUILIBRIUM VALUE

SINCE STABLE I USE FUT

$$V = \lim_{s \rightarrow 0} s G(s) \frac{1}{s} = \lim_{s \rightarrow 0} G(s) = G(0) = \frac{1}{10}$$

SETTLING TIME AT 5%.

STABLE, 2nd ORDER, STRICTLY PROPER

USE 2nd ORDER COMPLEX POLE

$$T_{d\epsilon} = -\frac{\ln \frac{0.01\epsilon}{\zeta \omega_n}}{\zeta \omega_n} = \frac{-\ln(0.01)5}{\frac{3}{4} \cdot 10} = 3.07 \text{ (s)} \quad \sqrt{\text{SECONDS}}$$

$$\epsilon = 5$$

$$\zeta = ?$$

$$\omega_n = ?$$

$$G(s) = \frac{N \omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2}$$

$$\omega_n^2 = 100 \Rightarrow \omega_n = 10$$

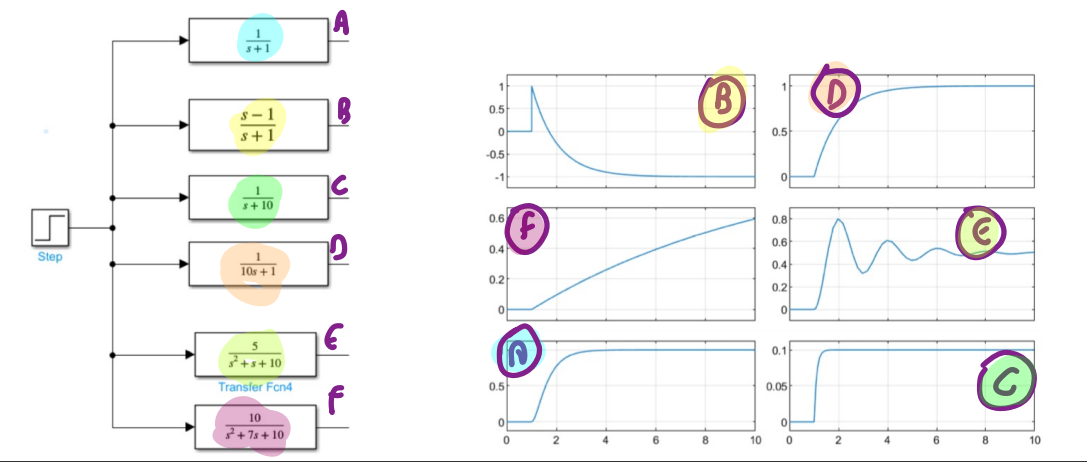
$$2\zeta \omega_n = 15$$

$$\zeta = \frac{15}{2} \cdot \frac{1}{\omega_n} = \frac{15}{20} = \frac{3}{4}$$

OVERSHOOT

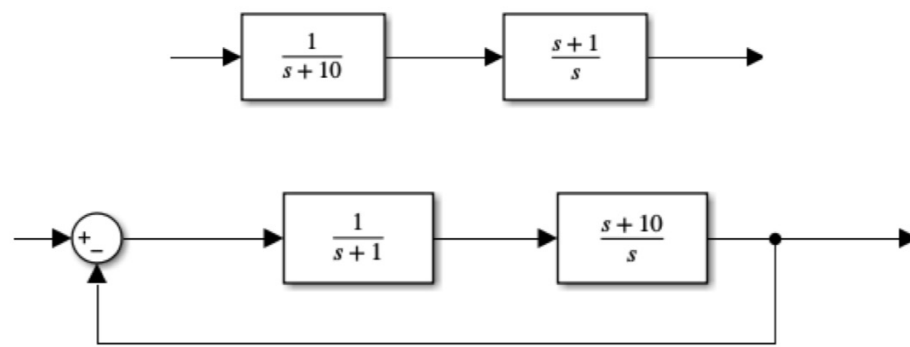
$$s\% = 100 e^{\frac{-\zeta \pi}{\sqrt{1-\zeta^2}}} = 100 e^{\frac{-\frac{3}{4} \pi}{\sqrt{1-9/16}}} = 2.3\%$$

Exercise 2)
Connect the systems with the correct step response

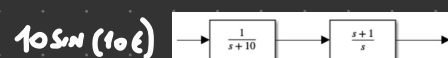


Exercise 3)

Given the systems



1. evaluate the response to the following signals:
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SYSTEM IN SERIES $\rightarrow$ MULTIPLY THE TRANSFER FUNCTION

$$G_{TOT}(s) = G_1(s) G_2(s) = \frac{1}{s+10} \cdot \frac{s+1}{s} = \frac{s+1}{s^2+10s}$$

1) FIND POLES

$$P = Z = \{0, -10\}$$

2) EQ. VALUE

$$N = \lim_{s \rightarrow 0} s \cdot G(s) = s \cdot \frac{s+1}{s(s+10)} = \frac{s+1}{s+10} = \frac{1}{10}$$

NOT $\frac{1}{s}$

$$Z = 1 = \{-1\}$$

$$q = P - Z = 2 - 1 = 1$$

2nd ORDER, STRATEGY PROBAB